AI Crypto Hedge Fund - Final Notebook

Execution mode: **FULL FINAL NOTEBOOK**.

This notebook is the single end-to-end reviewer narrative. It imports repository package code and reads committed validation/final-test artifacts. It does not place orders, download live data, call an external LLM or rerun/tune final-test strategies.

```
final_test_lock_sha256: c33b5eb396f60b1e2a7890616b8d9ae1cd69e91375dec925b68b6673d84
3af5e
final_test_exposure: EXPOSED
final_test_dir: artifacts/final_test/c33b5eb396f6
stage12_mode: FULL FINAL NOTEBOOK
```

1. Executive summary and coherent fund vision

The system is a reproducible historical research MVP for a risk-first AI-assisted crypto portfolio. Agents produce scored proposals with confidence and cutoffs; deterministic risk, allocation, rebalance and execution layers decide what can be simulated. The MVP is long-only, unlevered, daily spot and educational.

	Level	Selected	Net return	Benchmark	Sharpe	Max DD	Costs
0	level_1	SMA baseline	-7.4%	-5.4%	-0.171	-18.5%	$8,906
1	level_2	agent_ensemble	-0.6%	-5.4%	-0.522	-1.4%	$1,600
2	level_3	cvar_downside	-18.0%	-25.4%	-0.023	-45.2%	$1,493
3	level_4	calendar_monthly	-4.1%	-9.3%	-0.880	-9.1%	$3,584
4	level_5	large_universe_dynamic	-28.0%	-45.2%	-0.218	-42.2%	$110,939

2. Reproducibility/environment/data hashes

The final-test lock and artifact metadata identify data, config, git, costs, period, benchmark and seed values. The accepted lock is shown by package code above.

	Metric	Value
0	Data SHA-256	9f539f38394240f5...b80d7e14
1	Instruments SHA-256	df7777139dd41060...7bddf18b
2	Manifest SHA-256	24b2263f9ffd125f...37fcbd3e
3	Validation config SHA-256	da1dcaf442517b6c...e72e77e8
4	Generated final config SHA-256	c6c79b974e7c46f4...b1522fc6
5	Locked git commit	d200df6d8a5bd167...9943a1e5

6	Runner git commit	d200df6d8a5bd167...9943a1e5
7	Period	[2025-01-01, 2025-12-31]
8	Costs	{'chargeable_notional': 'risky_asset_notional_...

3. Data preparation, provenance and quality

The included dataset is frozen daily spot OHLCV with instrument metadata and a manifest. It is validated offline and carries the known active-market survivorship/delisting limitation.

	Metric	Value
0	Level 5 eligible pairs	120
1	Level 5 scored pairs	120
2	Level 5 selected holdings	25
3	System status	ok
4	Incident count	53
5	Fail-safe scenarios	["kill_switch_cash_schedule_unit_test", "volat...
6	Runtime	78.4s
7	Peak RSS	848 MiB

4. Architecture and agent interaction trace

The shared flow is: frozen data -> causal features -> typed agents -> aggregator -> pre-risk -> allocator -> rebalance controller -> post-risk -> orders/fills -> ledger -> metrics/monitoring. The next cell prints a committed end-to-end Level 2 trace.

	agent	symbol	score	confidence	fit_cutoff	feature_
0	sma_crossover	BTC/USDT	-1.000	0.228	2024-12-31T00:00:00+00:00	20 01T00:00:00-
1	econometric_ar_garch	BTC/USDT	0.037	0.049	2024-12-31T00:00:00+00:00	20 01T00:00:00-
2	ml_logistic	BTC/USDT	-0.128	0.128	2024-12-31T00:00:00+00:00	20 01T00:00:00-
3	ml_hist_gradient_boosting	BTC/USDT	-0.236	0.236	2024-12-31T00:00:00+00:00	20 01T00:00:00-
4	aggregator	BTC/USDT	-0.466	0.040	2024-12-31T00:00:00+00:00	20 01T00:00:00-
5	post_risk	portfolio	approve	cash=100.0%	2025-01-01T00:00:00+00:00	20 01T00:00:00-

5. Model validation and no-leakage protocol

Train data covers 2021-2023, validation covers 2024, and the frozen final test covers 2025. Stage 12 consumes the exposed final artifacts and does not retune choices.

6. Level 1 — Baseline Strategy for a Single Cryptocurrency.

BTC/USDT SMA baseline through the shared next-open engine.

	Net return	Sharpe	Max DD	Turnover	Costs	Benchmark
0	-7.4%	-0.171	-18.5%	598.7%	$8,906	-5.4%

7. Level 2 — Adding AI Agents, Econometrics and ML.

Single-pair technical, econometric, ML and ensemble agents.

	Approach	selected_for_level_2	Net return	Sharpe	Max DD	Turnover	Costs
0	technical_sma	no	-7.9%	-0.383	-18.9%	915.6%	$13,915
1	econometric_ar_garch	no	-3.7%	-1.411	-4.1%	2033.8%	$29,711
2	ml_logistic	no	-0.4%	-0.602	-0.6%	193.6%	$2,900
3	ml_hist_gradient_boosting	no	0.0%	0.020	-1.3%	462.3%	$6,947
4	agent_ensemble	yes	-0.6%	-0.522	-1.4%	106.5%	$1,600

8. Level 3 — Portfolio Management on Historical Data (5–7 assets, prior 12 months).

Static 5-7 asset portfolio estimated from the prior 12 months.

	Method	selected_for_level_3	Net return	Sharpe	Max DD	Turnover	Costs	Benchm
0	equal_weight	no	-25.4%	-0.125	-48.7%	99.5%	$1,492	-25
1	inverse_volatility	no	-16.4%	0.002	-44.8%	99.5%	$1,493	-25
2	minimum_variance	no	0.9%	0.280	-36.3%	99.5%	$1,493	-25
3	cvar_downside	yes	-18.0%	-0.023	-45.2%	99.5%	$1,493	-25

9. Level 4 — Dynamic Portfolio Rebalancing.

Dynamic small-portfolio rebalance policies selected on validation.

	Policy	selected_for_level_4	Net return	Sharpe	Max DD	Turnover	Costs	E
0	static_level3_benchmark	no	-18.0%	-0.023	-45.2%	99.5%	$1,493	
1	calendar_monthly	yes	-4.1%	-0.880	-9.1%	237.3%	$3,584	
2	drift_monthly	no	3.8%	0.290	-23.5%	839.0%	$12,733	
3	signal_risk_monthly	no	3.8%	0.290	-23.5%	839.0%	$12,733	

10. Level 5 — Portfolio Expansion to 100+ Pairs.

Large-universe point-in-time scoring with dynamic top-K allocation.

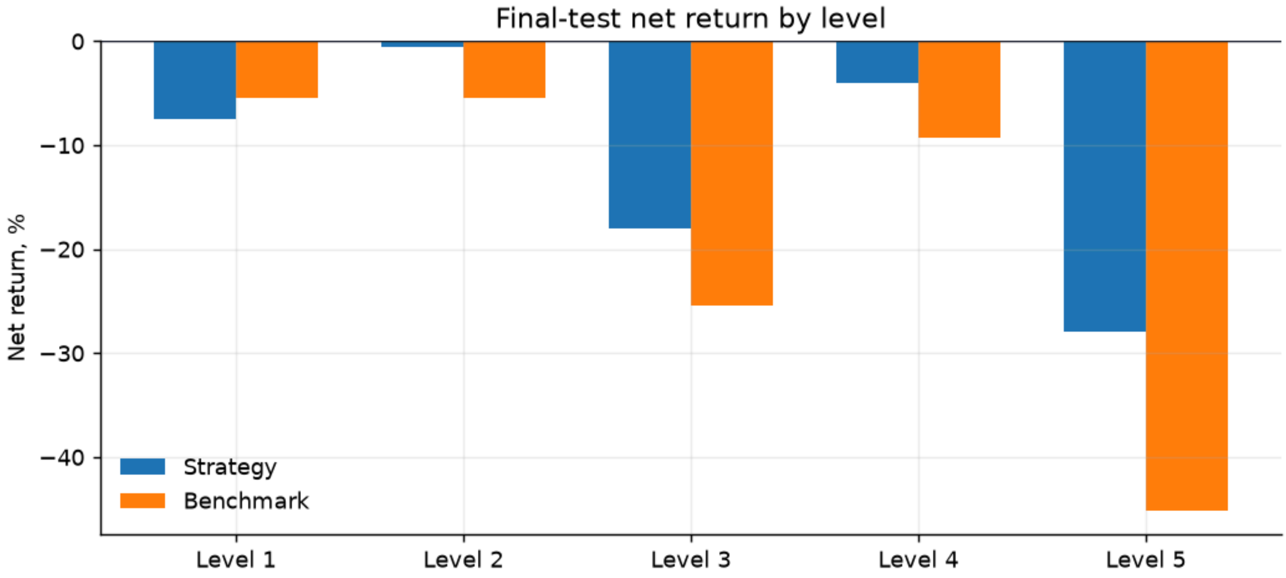
	Net return	Sharpe	Max DD	Turnover	Costs	Benchmark	Runtime	Peak RSS
0	-28.0%	-0.218	-42.2%	4924.5%	$110,939	-45.2%	78.4s	848 MiB

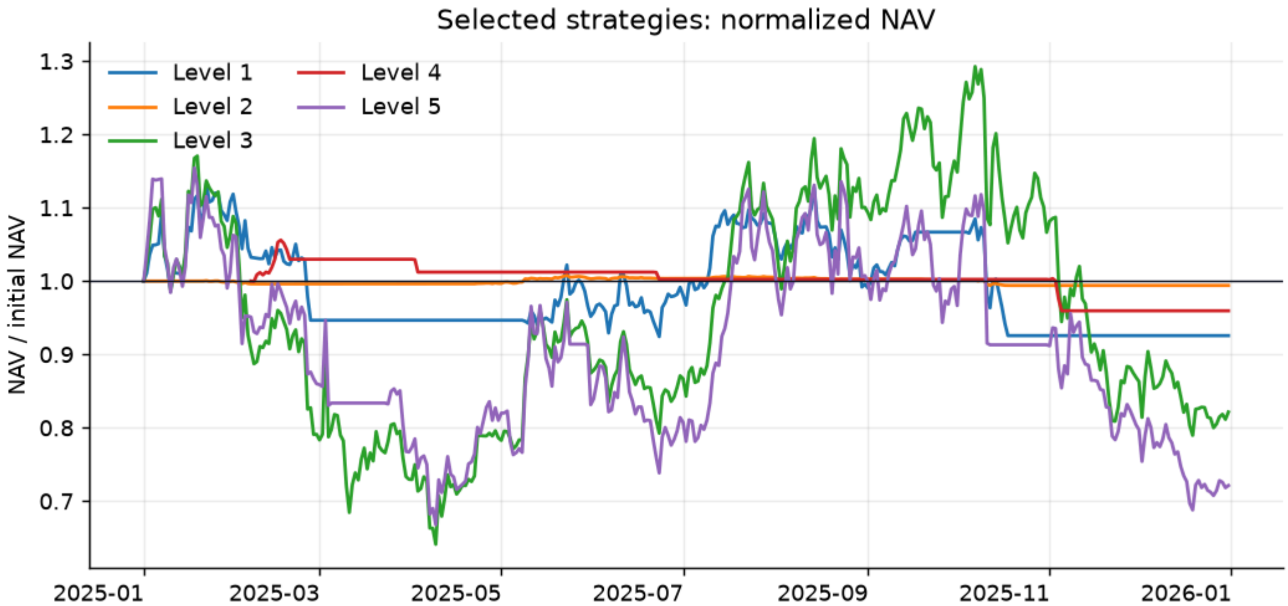
11. Cross-level comparison, monitoring and fail-safes

Net after fees/slippage is primary. Monitoring includes data/model/system health, agent disagreement, optimizer fallback, runtime, incidents and explicit fail-safe scenarios.

	Level	Selected	Net return	Benchmark	Sharpe	Max DD	Costs
0	level_1	SMA baseline	-7.4%	-5.4%	-0.171	-18.5%	$8,906
1	level_2	agent_ensemble	-0.6%	-5.4%	-0.522	-1.4%	$1,600
2	level_3	cvar_downside	-18.0%	-25.4%	-0.023	-45.2%	$1,493
3	level_4	calendar_monthly	-4.1%	-9.3%	-0.880	-9.1%	$3,584
4	level_5	large_universe_dynamic	-28.0%	-45.2%	-0.218	-42.2%	$110,939

	Metric	Value
0	Level 5 eligible pairs	120
1	Level 5 scored pairs	120
2	Level 5 selected holdings	25
3	Approved nonzero max	25





12. Limitations, real-trading application and production roadmap

Limitations: active-market survivorship/delisting bias, daily-bar liquidity proxies, USDT cash assumption, simplified fills, cash-heavy risk behavior, and Level 5 top-K benchmark rather than a full eligible-universe basket. Future work includes multi-CEX adapters, order-book liquidity, reconciliation, Telegram controls and news/sentiment ingestion. Those future items are not enabled in this MVP.